

PRODUCT REQUIREMENTS DOCUMENT

Transport & Logistics ERP System

East & Central Africa — Multi-Country Operations

This document is confidential and intended solely for the named client.

1. Document Purpose & Scope

This Product Requirements Document (PRD) has been prepared by Mori based on a thorough review of the Software Requirements Specification (SRS) provided by the client. Its purpose is to:

- Confirm mutual understanding of the full project scope before development accelerates
- Establish a clear, phased delivery plan across all 36 ERP modules
- Define non-negotiable technical and regulatory requirements (especially multi-country compliance)
- Serve as the contractual reference document for scope, timeline, and milestone approvals

The system is modeled after Microsoft Dynamics 365 Business Central and is designed to serve transport and logistics operations across six countries in East and Central Africa: Tanzania, Kenya, Uganda, Rwanda, Zambia, and the DRC.

2. Project Overview

2.1 Business Context

The client operates a cross-border transport and logistics business spanning two major trade corridors:

- Northern Corridor: Mombasa Port (Kenya) through Uganda to Rwanda and DRC
- Central Corridor: Dar es Salaam Port (Tanzania) through Zambia to DRC

The ERP system must handle the full business lifecycle — from freight quoting and trip dispatch to vehicle maintenance, regulatory compliance, multi-country payroll, and consolidated financial reporting — all within a single unified platform.

2.2 Key Business Challenges This ERP Solves

- Manual or disconnected tracking of vehicles, driver compliance documents, and border crossing requirements across 6 countries
- No unified financial visibility across subsidiaries operating in different currencies (USD, TZS, KES, UGX, RWF, ZMW)
- Difficulty tracking trip-level profitability (cost per truck, per corridor, per client contract)
- Risk of trucks being impounded at borders due to expired COMESA permits, health certificates, or insurance documents
- Cash disbursements to drivers via mobile money (M-Pesa, Airtel) are currently untracked in the ledger
- No real-time fuel efficiency monitoring to detect siphoning or mechanical issues

2.3 Compliance & Regulatory Environment

The system must comply with, and integrate with, the following country-specific regulatory requirements:

- Tanzania: TRA VFD (Virtual Fiscal Device) for e-invoicing; NSSF and PAYE payroll deductions
- Kenya: KRA TIMS for e-invoicing; SHIF (formerly NHIF) and PAYE payroll deductions; VAT at 16%
- Uganda / Rwanda / Zambia: Country-specific VAT rates and withholding tax rules (to be confirmed per jurisdiction)
- COMESA: Cross-border transit permits and Yellow Card insurance required for all vehicles crossing international boundaries

3. Phased Delivery Plan

Given the scale of this project (36 modules, 6 countries, 8 external API integrations), delivery is structured into 6 sequential phases. Each phase ends with a client review and formal sign-off before the next begins. The total estimated delivery timeline is 52 weeks (approximately 12 months).

Phase	Scope	Modules	Duration	Priority
Phase 1 Foundation	System Administration & Auth (RBAC, user roles, sessions) General Ledger & Chart of Accounts Basic Accounts Payable & Receivable Organization setup (multi-company, legal entities) Audit trail logging	M36, M8, M1, M2, M34, M31	Weeks 1–6	Critical
Phase 2 Core Finance	Budgeting & budget controls Cash & Bank Management M-Pesa / Airtel Money integration ledger Tax module (VAT/WHT per country) Multi-currency & exchange rate handling	M3, M4, M10, M5	Weeks 7–12	Critical
Phase 3 Operations	Fleet Management (vehicles, drivers, compliance docs) Transportation Management (load planning, routes, freight billing) Live Vehicle Tracking Map Common module (GPS waypoints, units of measure)	M11, M12, M30	Weeks 13–20	High
Phase 4 Inventory & Payroll	Inventory & Warehouse Management Procurement & Vendor Collaboration Portal Product Information Management Cost Management (FIFO/Weighted Avg) Multi-country Payroll (TZ/KE/UG/RW) Expense Management (driver cash sheets)	M13–18, M9, M23	Weeks 21–30	High
Phase 5 Assets & HR	Asset Management & Fixed Assets Depreciation calculations Production Control (workshop refits) Service Management (SLAs, breakdown dispatch) Human Resources directory Time & Attendance (biometric sync) Questionnaire / Safety checklists	M19–22, M24–26	Weeks 31–40	Medium
Phase 6 Commercial	Sales & Marketing (freight quotes) Retail / POS (fuel	M27–29, M5, M6, M7, M33,	Weeks 41–52	Medium

Phase	Scope	Modules	Duration	Priority
& Dashboard	stations, mobile money payments) Project Management & Accounting Consolidation module (multi-subsidiary rollup) Credit & Collections (aging buckets) Cost Accounting (per-truck, per-corridor P&L) Executive Dashboard & KPI cards Trip P&L reports Master Planning (truck deployment forecasting) Demo / Sandbox environment	M32, Dashboard		

4. Detailed Module Specifications

4.1 Core Financial Modules (M1–M10)

These modules form the accounting backbone of the entire ERP. No other module posts financial transactions independently — all monetary entries are routed through the General Ledger.

M1 — Accounts Payable

- Manages vendor master records with country-validated TIN and VAT registration numbers
- Supports payment terms (Net 30, Net 60, Cash on Delivery) and default currency per vendor
- Invoice matching engine links AP invoices to Purchase Orders with automated 3-way matching
- Withholding tax auto-calculated at invoice level based on service type and country of vendor

M2 — Accounts Receivable

- Customer registry with credit limit enforcement, credit days tracking, and tax exemption flags (diplomatic/transit)
- E-Invoicing integration: fiscal serial numbers generated via KRA TIMS / TRA VFD API at time of invoice posting
- Demurrage charges tracked per trip ID — critical for cross-border container delay billing
- Real-time current balance due as a calculated field

M3 — Budgeting

- CapEx and OpEx budget lines per cost center (e.g., Fleet Operations, Workshop Mombasa)
- Budget control behaviors: Strict Block (transaction rejected), Warning Only, or Director Override Required
- Real-time consumed amount updated on every financial posting

M4 — Cash & Bank Management

- Multi-bank account configuration: CRDB, KCB, Stanbic, and others with IBAN/SWIFT fields
- Mobile money integration ledger for M-Pesa and Airtel Money disbursements to drivers
- Each mobile transfer linked to a Driver ID, disbursement type (fuel allowance, tolls, border fees, emergency repair), and API callback status

M5 — Consolidations

- Mapping matrix: each subsidiary account (e.g., KE01) maps to parent holding ledger account (HQ01)

- Exchange rate types selectable per consolidation run: Spot Rate, Average Rate, or Historical Rate

M6 — Cost Accounting

- Profitability analysis per truck (Vehicle_ID), per client contract, and per corridor/lane
- Direct costs, allocated indirect costs, and revenue generated tracked per cost object
- Net margin percentage calculated automatically

M7 — Credit & Collections

- Aging bucket management: Current, 1–30 days, 31–60 days, 61–90 days, 90+ days overdue
- Collection statuses: Active, Under Dispute, Legal Action, Credit Blocked

M8 — General Ledger

- Standard chart of accounts structure with multi-currency revaluation
- Voucher-based posting with full financial dimension support (Branch: Dar HQ, Mombasa Branch, Kampala Office)
- Exchange rate fields recorded per posting for audit accuracy

M9 — Payroll

- Multi-country payroll: Tanzania (PAYE progressive brackets + NSSF) and Kenya (PAYE + SHIF) computed separately per employee
- Net salary calculated after all deductions; disbursement routed through Cash & Bank module

M10 — Tax

- Tax component codes per country: TZ-VAT-18 (18%), KE-VAT-16 (16%), and others
- Tax payable and tax recoverable accounts linked per tax component
- Automated tax return preparation data per filing period

4.2 Operations & Logistics Modules (M11–M12)

M11 — Fleet Management

- Vehicle registry: VIN/chassis, license plate (Tanzanian T-series and Kenyan KAA-series formats), make/model (Scania, Volvo, Shacman)
- Fuel efficiency targets: KM per liter (loaded) set per vehicle for efficiency monitoring
- Driver compliance records: license number and expiry, COMESA transit permit, passport expiry, Yellow Fever certificate status
- Automated warning flags when any compliance document approaches expiry (configurable threshold, minimum 14 days)

M12 — Transportation Management

- Load planning workbench: gross weight (KG) and volume (CBM) per load
- Route engine: corridor type (Northern or Central), mileage, transit time hours, and zone master
- Freight bill management: reconciliation status (Unreconciled, Matched, Discrepancy) against carrier invoices
- Trip P&L report: revenue minus all direct expenses (fuel, tolls, driver allowances, demurrage) per trip

4.3 Inventory & Supply Chain Modules (M13–M18)

- Inventory covers: truck spare parts (e.g., Brake Pad Scania Front), tyres (serialized tracking), lubricants (batch tracking), and bulk fuel reserves
- Costing methodology per item group: FIFO, Weighted Average, or Standard Cost
- Minimum reorder levels trigger procurement alerts automatically
- Vendor Collaboration Portal: external vendor access to open POs, bid submission, and invoice uploads
- Warehouse Management: bin-level spatial tracking, storage temperature settings (Ambient, Chilled, Frozen)

4.4 Asset Maintenance & Workshop Modules (M19–M22)

- Fixed asset registry for non-fleet infrastructure: workshop machinery, cranes, depot equipment
- Asset condition scoring (1–10), maintenance checklists, and accumulated maintenance cost tracking
- Depreciation profiles: Straight Line or Reducing Balance, with configurable useful lifecycle months
- Production Control: workshop re-manufacturing and trailer axle refit order scheduling with estimated vs. actual hours
- Service Management: SLA response time targets, external client truck maintenance contracts, and emergency breakdown dispatch

4.5 Human Capital Modules (M23–M26)

- Expense Management: driver trip cash sheets with receipt image attachments; categories include border fees, weighbridge penalties, out-of-pocket spares, and driver meals
- HR directory: role-based (Longhaul Driver, Workshop Mechanic, Depot Clerk, Accountant), employment status tracking
- Safety questionnaires and post-trip security checklists for drivers and workshop staff
- Time & Attendance: biometric terminal sync for clock-in/out with total hours calculated automatically

4.6 Commercial & Retail Modules (M27–M29)

- Sales & Marketing: freight quotation lifecycle (Draft, Sent, Accepted, Expired, Lost) with origin/destination zone and quoted rate
- Retail / POS: point-of-sale logs for fuel stations and convenience stores; payment modes include Cash, Card, Mobile Money, and Fuel Card
- Project Management & Accounting: high-value infrastructure and pipeline logistics projects with contract value, budget limits, and stage tracking

4.7 System & Shared Framework Modules (M30–M36)

- Common module: global GPS waypoint registry (named checkpoints: Chalinze, Salama Weighbridge, Nakuru Stop, etc.) with lat/long coordinates
- Audit Workbench: every field modification logged with old value, new value, user ID, and IP address — immutable records
- Master Planning: truck deployment forecasting — available trucks vs. required trucks with variance/shortage flag
- Organization Administration: multi-company legal entity setup with country, fiscal calendar, and Company_ID (4-character, e.g., TLOG)
- System Administration: RBAC configuration, user account management, account locking, printing endpoint setup
- Demo / Sandbox: isolated database partition for testing without affecting production data

5. External API & Integration Requirements

The following external integrations are required by the system. All integrations must be handled with proper error handling, retry logic, and API execution status tracking (Success, Failed, Timeout, Pending Verification).

Integration	Country / Provider	Requirement	Phase
KRA iTax / TIMS (e-invoice)	Kenya	API response ≤ 3 seconds	Phase 2
TRA VFD (fiscal device)	Tanzania	API response ≤ 3 seconds	Phase 2
M-Pesa API (Safaricom)	Kenya / Tanzania	Disbursements to drivers with callback tracking	Phase 2
Airtel Money API	Uganda / Rwanda / Zambia	Disbursements with callback tracking	Phase 2
Biometric terminal sync	All depots	Clock-in/out records auto-import	Phase 5
GPS live vehicle tracking	All corridors	Real-time map, panic alerts, route deviation	Phase 3
CRDB / KCB / Stanbic bank feeds	TZ / KE / Multi-country	Statement reconciliation	Phase 2

6. Security & Access Control

6.1 Role-Based Access Control (RBAC)

The system enforces strict role segregation. No user may be assigned conflicting roles (e.g., Finance Controller and Operations Planner simultaneously) without System Administrator approval.

Role	Module Access	Key Restrictions
System Administrator	Full access — all 36 modules, database logs, sandbox	None — full system authority
Finance Controller	General Ledger, AP, AR, Budgeting, Tax, Consolidations, Payroll	No fleet, transport, or HR operational access
Operations Planner	Fleet Management, Transport Management, Common, Sales & Marketing	No financial posting or payroll access
Workshop Manager	Asset Management, Inventory Management (spares), Time & Attendance	Read-only on financial figures
Driver (Mobile App)	Expense submissions, trip check-ins, document uploads	Offline-first mobile; no desktop system access

6.2 Data Partitioning

- Each legal entity's data is fully partitioned by Data_Area_ID — a Finance Controller in KE01 cannot view TZ01 financial data unless explicitly granted cross-entity access
- All cross-entity consolidation operations are logged in the Audit Workbench
- Mobile app users (drivers) operate in a restricted environment with no access to financial, HR, or system configuration data

7. Non-Functional Requirements

Category	Requirement	Target
Tax API Latency	KRA TIMS and TRA VFD e-invoice API handshake must complete from ledger submission trigger	≤ 3 seconds
Offline Resilience	Field mobile app must cache expense entries, weighbridge records, and delivery checkpoints locally; auto-sync on network restore	Full offline capability
Multi-currency	All financial modules must support USD, TZS, KES, UGX, RWF, ZMW simultaneously with configurable exchange rate types (Spot, Average, Historical)	6 currencies
Multi-company	Each legal entity (TZ01, KE01, UG01, HQ01, etc.) must be fully data-partitioned; consolidation module aggregates into parent ledger	6+ entities
Document Expiry	System must flag vehicle and driver compliance docs (COMESA, insurance, passports, health certificates) expiring within configurable warning windows	14-day warning
Audit Trail	Every record modification must log: User ID, old value, new value, timestamp, and IP address of origin — non-deletable	100% coverage
GUID Global Fields	Every master record and transaction must inherit: Record_ID (GUID), Data_Area_ID, Created_By, Created_Timestamp, Last_Modified_By, Last_Modified_Timestamp, Row_Version	All modules

7.1 Global Metadata Schema (Mandatory for All Records)

Every master record and transaction ledger entry across all 36 modules must inherit the following fields. This is a hard architectural requirement, not optional:

- Record_ID — GUID, system-generated Primary Key
- Data_Area_ID — Legal entity identifier (e.g., TZ01, KE01, HQ01)
- Created_By_User_ID — User login string
- Created_Timestamp — DateTime, immutable after creation
- Last_Modified_By_User_ID — User login string
- Last_Modified_Timestamp — DateTime, updated on every save
- Row_Version — Timestamp / concurrency hash to prevent conflicting simultaneous edits

8. Executive Dashboard Requirements

8.1 KPI Cards (Top Row)

- Active Vehicles / Total Fleet — trucks on active trips vs. in yard or maintenance
- Active Drivers / Total Personnel — dispatched vs. on rest cycle (driver fatigue management compliance)
- Asset Utilization Rate (%) — percentage of fleet generating revenue today
- Total Bookings & Total Booking Value — cargo contracts in current period
- Invoices Issued vs. Payments Received — real-time liquidity indicator
- Cash in Hand & Bank Balances — aggregated across all regional banking accounts
- Supplier Invoices vs. Payments Made — outstanding payables to carriers and fuel suppliers

8.2 Charts & Gauges (Middle Row)

- Customer Payment Recovery Gauge — half-donut: Received (green) vs. Pending/Overdue (red)
- Supplier Obligations Gauge — half-donut: Paid (green) vs. Outstanding (red)
- Fuel Efficiency Monitor — bar chart per vehicle: Meets Target (green) / Below Expected (red) / No Data (grey)

8.3 Compliance & Lifecycle Tracking (Bottom Row)

- Vehicle Maintenance Status — arc chart: Completed / Upcoming / Due Soon / Overdue
- Regulatory Document Lifecycles — table showing Current (green) / Expiring Soon (yellow) / Overdue/Expired (red) for both vehicles and drivers
- Border Turnaround Tracker — live average delay hours at major crossing points (Tunduma, Malaba, Namanga)
- Demurrage Warning Bar — financial alerts when containers approach free-period deadlines
- Live Transit Incidents Feed — signal dropouts, route deviations, panic button alerts
- Trip P&L Report — per-trip profit and loss: revenue minus all expenses

9. Assumptions, Dependencies & Open Items

9.1 Assumptions

- The client will provide access credentials and sandbox/test accounts for all government tax APIs (KRA TIMS, TRA VFD) prior to Phase 2 integration work
- The client will provide sample bank statement formats from CRDB, KCB, and Stanbic for reconciliation mapping
- GPS hardware/telematics devices are already installed in vehicles or will be procured independently by the client; this ERP connects to the data feed via API
- Biometric attendance terminals are already installed or will be installed by the client; this ERP consumes the output data
- Country-specific tax rates for Uganda, Rwanda, and Zambia will be confirmed by the client's local accountants before Phase 2 begins
- All development, review, and feedback cycles will be conducted in English

9.2 Out of Scope

- Procurement and installation of GPS hardware or biometric devices
- Custom integrations with government tax portals beyond KRA TIMS and TRA VFD (unless separately agreed)
- Mobile app development for Android/iOS (the mobile-responsive web app is in scope; a native app is a separate engagement)
- Historical data migration from existing systems (migration plan to be scoped separately)
- Ongoing post-launch support and SLA maintenance (to be agreed under a separate support contract)

9.3 Open Items Requiring Client Confirmation

#	Open Item	Owner	Required By
1	Confirm VAT/WHT rates for Uganda, Rwanda, and Zambia	Client (local accountants)	Phase 2 start
2	Provide KRA TIMS and TRA VFD sandbox API credentials for integration testing	Client	Phase 2 start
3	Confirm list of legal entity Company IDs (TZ01, KE01, HQ01, etc.)	Client	Phase 1 start
4	Confirm GPS telematics provider and API documentation for vehicle tracking feed	Client	Phase 3 start
5	Confirm biometric attendance terminal brand/model and export format	Client	Phase 5 start
6	Confirm chart of accounts structure or provide existing account codes to map	Client	Phase 1 start

Thank you for your continued partnership. We look forward to building this system together.